Duke University Science Olympiad
Water Quality
January 30, 2021

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Rules

1. You have 50 minutes to complete this exam. This will be monitored by timestamps on the Scilympiad platform.
2. You may not discuss the exam with anyone except your partner until 9:01 AM on January 31, 2021.
3. This exam is split into three parts. Tiebreakers are marked with ***.
4. You may use the "classroom" feature on Scilympiad to ask questions.

Honor Statement

By signing below, I affirm that:

- I will adhere to the Science Olympiad Code of Ethics and General Rules.
- The tools I will use are within the parameters of this event.
- I will not communicate with anyone other than my partner during this test.
- I will not use the internet to help me take this test.
- I will not share any questions or answers on this test to other participants of this event.
- I will complete this event within the allocated time.

Signature _____

Signature _____

I. Marine and Estuary Ecology

You are a toxicologist, and you get called to a small town in eastern North Carolina where some residents have fallen ill after drinking the municipal water. Upon arrival, you learn that the town gets its water from the Cape Fear River watershed, which is contaminated with GenX and other forms of per- and polyfluoroalkyl substances (PFAS). These chemicals are present in the drinking water of an estimated 1 million North Carolinians.

1. Based on what you know about PFAS, list three symptoms the residents might exhibit. (3)

2. Some pregnant people exposed to GenX were concerned about it harming their unborn children. How might unborn babies be exposed to these chemicals? (2)

3. A study conducted by NC State University found that GenX does not stay in the bloodstream for an extended period of time. Despite this, GenX was shown to have detrimental effects on the offspring of pregnant mice. Come up with an explanation for how this occurs. (4) ***

4. Which bond prevents PFAS chemicals from degrading in the environment? (1)
    a. Carbon-fluorine
    b. Carbon-hydrogen
    c. Carbon-carbon
    d. Carbon-chlorine

5. Which of the following are PFAS chemicals? (3)
    a. Pentafluoroethane
    b. Chloropentafluoroethane
    c. 1,2-Dibromotetrafluoroethane
    d. 1-Butene
    e. 1,4-Dichloropentane
    f. Ethanol

6. Aside from providing drinking water, watersheds have other functions. List two of them. (2)

7. How does nonpoint-source pollution occur? (2)

II. Coral Reef Macroflora and Fauna

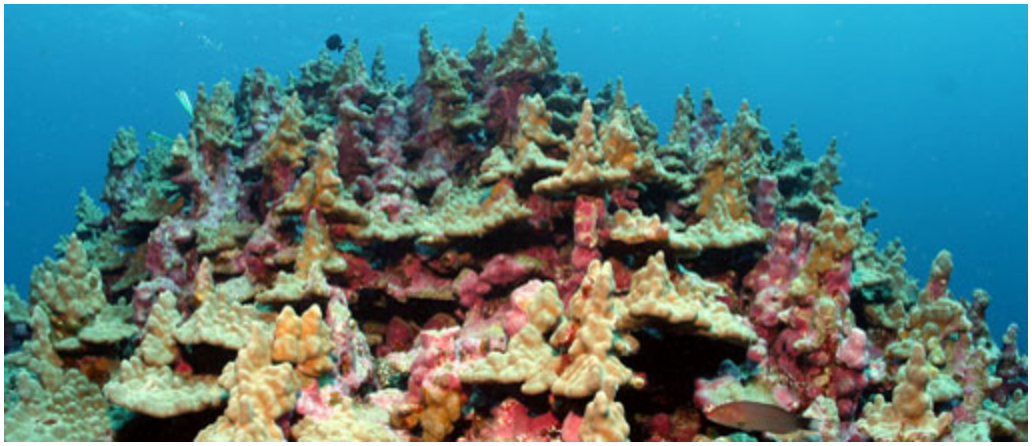


Image via Coral Health Atlas (hawaii.edu)

This is an image of Neva Shoals, a coral reef near Lisianski Island in Hawai'i. It covers 241,916 acres, nearly the size of O'ahu. (For comparison, the Great Barrier Reef is about 800,000 acres long.)

8. Which type of coral forms reefs? (1)
    a. Hermatypic
    b. Ahermatypic

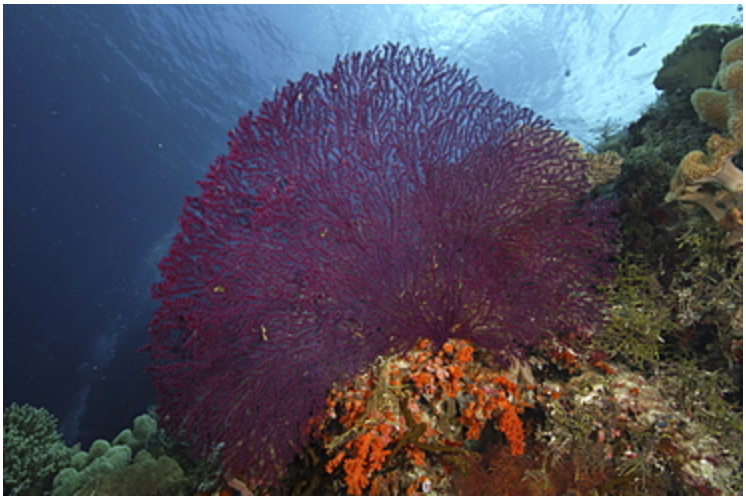
9. According to data compiled by the University of Hawaii, macroalgal overgrowth was the most prevalent disease that impacted Neva Shoals in 2012. What causes this condition, what are some indicators of it, and what are the consequences? (4)

10. Although macroalgal overgrowth was more prevalent, bleaching was more severe. What causes bleaching, what are the indicators, and what are the consequences? (4)

11. What percentage of marine life do coral reefs provide a habitat for? (2)

12. Explain the life cycle and reproductive process of coral reefs. Remember to explain what each step of reproduction entails. (10) ***

13. You're traveling in the Indo-Pacific region and come across a reef that has turned partially white. You see some sea cucumber mulling about, but you also notice that there are no butterflyfish to be found. What does this indicate? (5)



Above is an image of an organism found primarily in the Atlantic region of the world. In recent years, this organism has come to be recognized as part of an important foundational species of deep-sea benthic ecosystems. Refer to this as organism A.

14. What is the common name of organism A? (1)

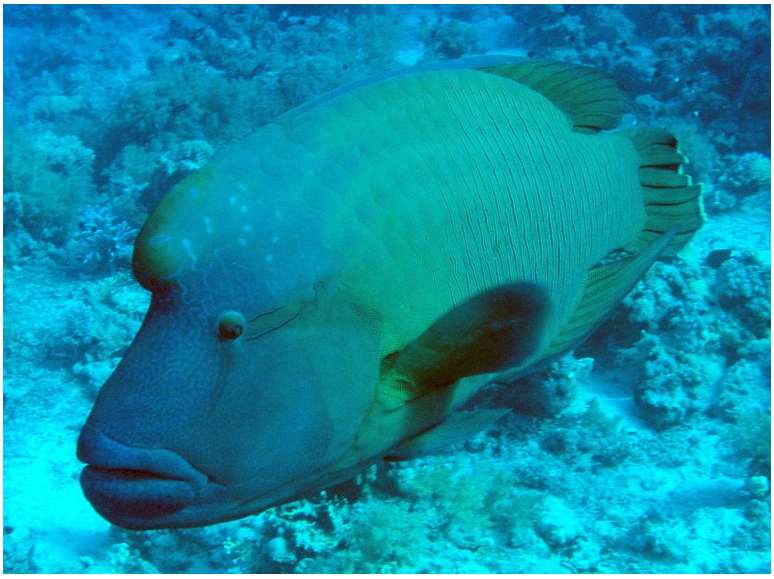
15. Which color does organism A not naturally appear in? (1)

16. Because organism A is a filter-feeder, it thrives in environments where the water flow is _____? (1)

17. True or False: greater than 1 million zooxanthellae are located in a single square millimeter of organism A's tissue? (1)

18. Briefly describe the sexual reproductive method(s) of organism A? (4)

19. When does organism A use sexual reproduction vs. when does it use asexual reproduction? (2)



Above is an image of an organism found primarily in the Indo-Pacific region of the world, often called “the king of the reef”. As of 2004, this organism is listed as endangered on the IUCN Red List. Refer to this as organism B.

20. What is the common name of organism B? (1)

21. What specific type of hermaphrodite is organism B classified as? Explain what this means. (2)

22. Briefly describe what hermaphroditism is. Name and describe another type. (4)

23. True or false: Organism B consumes both mollusks and sea urchins? (1)

24. How long can organism B grow, in feet? (1)

25. How can you tell a juvenile of organism B apart from an adult appearance-wise? (2)

The following questions are NOT associated with a specific image:

26. True or false: Groupers reproduce in spawning aggregations. (1)

27. True or false: Butterfly fish thrive in basic water. (1)

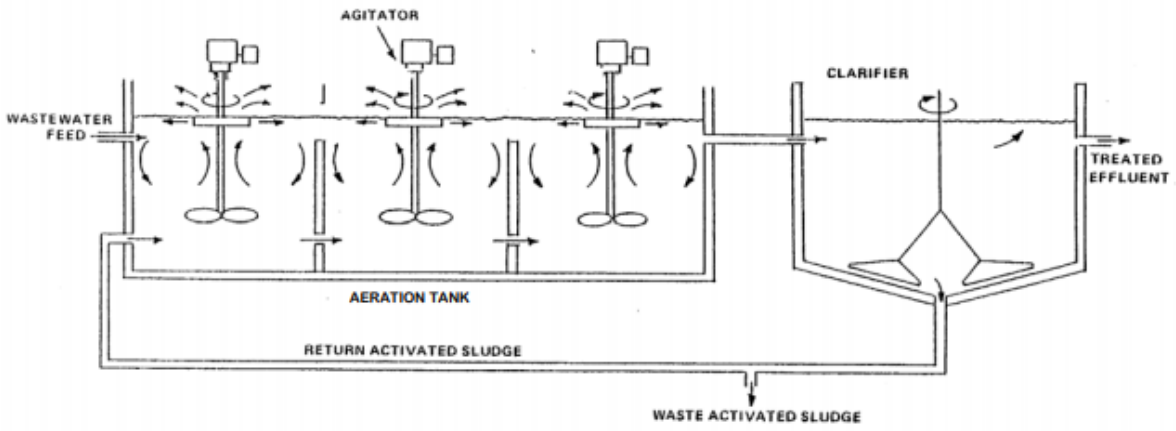
28. True or false: Lobsters typically eat the skin they have shed from molting. (1)

29. True or false: Male banded coral shrimp are typically larger than females. (1)

30. True or false: Fleshy algae typically have a canopy height of greater than 1 cm. (1)

III. Water Monitoring and Analysis

31. You are starting your day at the wastewater treatment plant where you work. You take a glance at the readings and see that this sample of water has a high biochemical oxygen demand index. What does this indicate? (3)



32. Explain the process depicted in the diagram above. (5) ***



Above is an image of a USGS hydrographer collecting a suspended-sediment water sample from the Little Colorado River, Grand Canyon, Arizona.

33. What is/are the difference(s) between the transmission light and scattering light methods of detecting turbidity? (4)

34. True or false: Is turbidity a direct indicator of health risk? Explain. (3)

35. The water in this image is visibly cloudy, so it must be at a level greater than __ NTU. (1)

36. What is the relationship between turbidity and water temperature? Why? (3)

The following questions are NOT associated with a specific image:

37. True or false: If the Secchi depth measured is 7 meters, the range of approximate depth that light can penetrate into the body of water is also 7 meters. (1)

38. True or false: A Water Quality Index (WQI) value of 12 represents excellent water quality. (1)

39. True or false: There is an inverse relationship between temperature & biological oxygen demand. (1)

40. True or false: Oxidation reduces the biological oxygen demand of wastewater. (1)

41. What is the major limiting nutrient in marine habitats? (2)